A FILTRATION AS IN PROPERTY (P) GIVES AN ADMISSIBLE SHORT EXACT SEQUENCE ON THE DIRECT SUM OF ITS STEPS

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-property-P-sequence (source lines 2424-2436); the proof environment begins two lines below it and is its own proof, closing at source line 2448.

Editorial note. No \ref or \cite occurs anywhere inside the retained span, so the package declares neither a source pointer nor a bibliography entry; property (P) is the author's own surrounding definition in the same chapter.

Editorial note. The statement and the proof each display a direct sum; both are part of the retained source range and are reproduced as written, so no figure file is needed.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-dga/preamble.tex, 217 lines), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Differential Graded Algebra chapter, counted once: a property (P) filtration turns the direct sum of its steps into an admissible short exact sequence mapping onto the module.

Allowed standard prerequisites (editorial).

- Filtration as in property (P). An exhaustive increasing filtration of P by dg submodules whose successive quotients are direct sums of shifted copies of A.. It supplies the successive inclusions F_i P -> F_{i+1} P that the construction of the two maps uses (source lines 2426-2432, 2440-2446).
- Admissible short exact sequence of dg modules. A short exact sequence of dg modules which splits as graded modules, so that the middle term is the direct sum of the outer terms.. The conclusion is precisely that this sequence is admissible, which is proved by writing down the splitting (source lines 2434-2436, 2446-2448).
- Splitting of the successive inclusions. For each i the inclusion F_i P -> F_{i+1} P admits a graded left inverse s_i.. The composed maps s_{j,i} and the tuple formula for the left inverse are built from those s_i (source lines 2444-2448).

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Lemma 0.1. *Let (A, d) be a differential graded algebra. Let P be a differential graded A-module. If $F_\bullet$ is a filtration as in property (P), then we obtain an admissible short exact sequence*

$$0 \to \bigoplus F_i P \to \bigoplus F_i P \to P \to 0$$

of differential graded A-modules.

Proof. The second map is the direct sum of the inclusion maps. The first map on the summand $F_i P$ of the source is the sum of the identity $F_i P \to F_i P$ and the negative of the inclusion map $F_i P \to F_{i+1} P$. Choose homomorphisms $s_i : F_{i+1} P \to F_i P$ of graded A-modules which are left inverse to the inclusion maps. Composing gives maps $s_{j,i} : F_j P \to F_i P$ for all $j > i$. Then a left inverse of the first arrow maps $x \in F_j P$ to $(s_{j,0}(x), s_{j,1}(x), \ldots, s_{j,j-1}(x), 0, \ldots)$ in $\bigoplus F_i P$. $\square$